A Hybrid Smishing Detection Framework: Integrating Machine Learning, Rule-Based Filtering, and Domain Reputation Analysis

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***Abstract*—The rise of SMS phishing (smishing) attacks poses a significant threat to mobile security, particularly for Vietnamese users whose linguistic environment is inadequately served by existing international solutions. This paper makes three primary contributions. First, we construct a Vietnamese smishing dataset comprising 2,600 labeled SMS messages, collected from real-world devices and community-sourced reports. Second, we propose a Hybrid Smishing Detection Framework featuring a four-layer linguistic preprocessing pipeline designed to handle Vietnamese text obfuscation, including teencode, Leetspeak, and broken Telex input. Third, we integrate this pipeline with a tree-based ML classifier and a Hybrid Decision Logic that incorporates Domain Verification. On a held-out test set of 520 samples, the XGBoost-based system achieves an overall accuracy of 97.5% and a smishing recall of 78%, outperforming single-model baselines in the precision-recall trade-off relevant to cyber-security deployment.**

***Index Terms*—Smishing, SMS Phishing, Machine Learning, Vietnamese NLP, Linguistic Feature Extraction, Hybrid Architecture**

I. Introduction

In the modern digital era, SMS remains one of the most reliable communication channels, with an open rate and near-instantaneous readership that far surpass those of other channels such as email. This trust has made it a prime vector for *smishing* (SMS Phishing) - a form of social engineering where attackers deceive victims into revealing sensitive information or installing malware. In Vietnam, attackers routinely impersonate banks, e-commerce platforms, and government agencies, often exploiting Brandname abuse to bypass traditional filters, causing significant financial and data losses for unsuspecting users.

Existing defenses remain limited: static blacklists and simple keyword matching fail against obfuscated text (teencode, Leetspeak, mixed-character URLs) and dynamic URL shorteners. Advanced ML pipelines from the international literature are furthermore optimized for English; applying them to Vietnamese yields suboptimal results due to diacritical marks, tone-dependent semantics, and local obfuscation conventions. To the best of our knowledge, no comprehensive system currently adapts these architectures specifically for the Vietnamese linguistic and telecommunications landscape.

Formally, smishing detection is framed as a binary text classification task. Given an incoming SMS message T with sender type S, the system must assign a label $L \in \{\textit{Smishing}, \textit{Legitimate}\}$. Specifically:

- **Input:** A tuple (T, S), where T is the raw SMS message body and S denotes the sender type (e.g. Brandname, Phone Number, Unknown).
- **Output:** A label L, a confidence score Score $\in [0, 1]$, and a natural-language explanation of the decision.

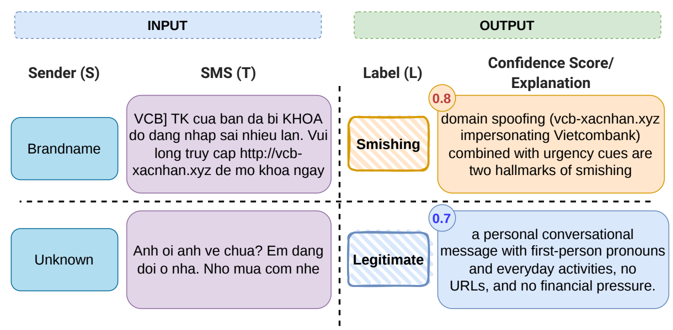

Fig. 1. Examples of each type of SMS.

To address the challenges above, we propose a Hybrid Real-Time Smishing Detection System tailored for Vietnamese. Our key contributions are:

1) **Dataset:** A composite Vietnamese dataset of 2,600 labeled SMS samples built via dual-channel acquisition and a five-annotator majority-vote protocol.
2) **Framework:** A four-layer linguistic preprocessing pipeline for Vietnamese, combined with an ML classifier and deterministic Domain Verification integrated through a Hybrid Decision Logic.
3) **Experiments:** Comparative evaluation of Decision Tree, Random Forest, and XGBoost demonstrating our hybrid approach outperforms TF-IDF baselines in the precision-recall trade-off critical for cyber-security deployment.

The remainder of this paper is organized as follows. Section II surveys related smishing detection methods and publicly available datasets, motivating our Vietnamese-specific approach. Section III describes the dual-channel data collection strategy and annotation protocol used to construct our 2,600-sample corpus. Section IV details the four-layer linguistic preprocessing pipeline designed to handle Vietnamese text obfuscation. Section V presents the proposed Hybrid Smishing Detection Framework, encompassing the ML classifier, Domain Verification module, and Hybrid Decision Logic. Experimental results and comparative evaluation against baseline models are reported in Section VI, followed by an error analysis in Section VII. Section VIII concludes the paper and outlines directions for future work.

II. Related Work

A. Related Methods

The landscape of smishing detection has evolved from simple rule-based filters toward increasingly sophisticated machine learning and deep learning approaches. Broad surveys confirm that machine learning is among the most promising anti-smishing methods, yet continuous evaluation of algorithms is required as attacker tactics evolve [1]. Within this space, deep learning architectures have attracted significant attention: Mahmud *et al.* [2] propose a hybrid model combining Bidirectional Gated Recurrent Units (Bi-GRUs) with Convolutional Neural Networks (CNNs) for smishing classification, while Mehmood *et al.* [3] aggregate CNNs and Long Short-Term Memory (LSTM) networks to automatically extract features from SMS text and reduce false-positive rates. Beyond text-only input, Shinde *et al.* [4] expand the feature space using optical character recognition (OCR) to detect scams embedded in image-based messages, combining unsupervised and deep semi-supervised learning.

Despite their expressive power, deep architectures demand large labeled corpora and sacrifice interpretability-two critical constraints in our setting. Mishra and Soni [5] demonstrate that a Defense-in-Depth hybrid coupling a Domain Checking Phase (URL validation via search-engine signatures) with an SMS Classification Phase based on five hand-crafted features achieves 97.93% accuracy while remaining fully interpretable. Goel *et al.* [6] further show that explicit text normalization before classification yields 96.2% accuracy on an English dataset; we extend this normalization philosophy to Vietnamese, where tone-mark interactions, Telex artifacts, and Leet obfuscation demand a multi-layer pipeline rather than a single normalization pass. Mishra and Soni also found that 74.22% of smishing messages contain character substitutions (e.g. “P@ytm”→“Paytm”) to bypass keyword filters, a finding corroborated by Li and Zeng [7] who document how such substitutions systematically evade dictionary-based detectors-a challenge amplified in Vietnamese due to diacritical marks. Crucially, Tuan *et al.* [8] evaluate SVM, Naïve Bayes, Random Forests, CNN, and LSTM specifically on Vietnamese SMS, noting measurable performance differences when processing text with and without diacritics and confirming that language-specific treatment is indispensable. Vietnamese noisy text modeling has also been shown to be important for mining informal online content [9]. Building on these findings, we adopt XGBoost [10] - a scalable tree-boosting system with a sparsity-aware split-finding algorithm and second-order gradient statistics-as our core classifier, since the 28-dimensional combined feature vector (27 linguistic features + sender type) constitutes exactly the kind of structured, heterogeneous input for which gradient-boosted trees excel over both shallow models and sparse TF-IDF representations, while remaining interpretable and efficient for real-time deployment.

B. Related Datasets

A persistent challenge in smishing research is the scarcity of labeled, publicly available datasets that reflect real-world attack patterns. Almeida *et al.* [11] introduced the UCI SMS Spam Collection, comprising 5,574 English-language messages, which became the de facto benchmark for SMS spam research; however, it does not capture the linguistic properties of other languages. More recently, Timko and Rahman [12] compiled a large community-sourced smishing corpus from Smishtank.com, providing detailed metadata including sender information, message bodies, and referenced brands-advancing the field with richer annotation but remaining English-centric.

On the Vietnamese side, Nguyen *et al.* [13] developed ViLexNorm, a lexical normalization corpus of annotated Vietnamese social media sentence pairs designed to address intended and unintended linguistic variations such as teencode and Leetspeak. While ViLexNorm provides a valuable normalization resource, no publicly available Vietnamese-specific smishing dataset exists. Our work addresses this gap by constructing a corpus of 2,600 labeled SMS messages through a dual-channel acquisition strategy, enabling evaluation under Vietnamese obfuscation conventions and Brandname-abuse patterns that no existing public dataset covers.

III. Data Collection

Due to the scarcity of standardized Vietnamese smishing datasets, we constructed a composite corpus through a dual-channel acquisition strategy.

a) Manual Transcription from Public Sources.: The first channel harvested confirmed smishing messages frequently reported as screenshots on Vietnamese cyber-security forums and anti-scam community groups. We performed manual transcription rather than OCR to accurately preserve linguistic nuances - teencode, special character obfuscations, and malicious URLs - that automated recognition tools often corrupt. This yielded a high-quality set of distinct smishing signatures targeting banking and e-commerce sectors.

b) Real-World Data Acquisition via Mobile Utility.: The second channel collected real-world SMS data - legitimate conversations, OTPs, and carrier advertisements - from real Android devices using *SMS Backup & Restore*, with explicit consent from participants. This application exports local message repositories into a structured XML format, allowing efficient extraction of message body, timestamp, and sender information.

c) Dataset Summary and Annotation Protocol.: Following aggregation, cleaning, and normalization, the final dataset comprises 2,600 unique rows (Table I). Labels were assigned by five annotators using pre-defined guidelines: a message is *Smishing* if it contains (1) a URL leading to a non-whitelisted or suspicious domain, (2) impersonation of a trusted entity, (3) financial pressure or urgency combined with a request for personal information, or (4) unrealistic recruitment offers. Final labels were decided by majority vote; unresolved samples were discarded to maintain dataset purity.

TABLE I
DATASET CLASS DISTRIBUTION

Class	Count	Percentage
Legitimate (Ham)	2,353	90.5%
Smishing (Spam)	247	9.5%
Total	**2,600**	**100%**

IV. DATA PREPROCESSING

SMS data contains significant noise and lacks structural uniformity. In spam messages, obfuscation techniques-inserting random spaces within URL segments, applying Leetspeak, or mixing characters-actively bypass keyword-based filters. A single-pass normalization approach fails to address these interacting noise sources, motivating our four-layer design. Each layer targets a distinct noise category; their outputs also serve as discriminative classifier features.

a) Layer 1 - Entity Masking.: Specific entities are replaced with standardized tokens (e.g. `<URL>`, `<PHONE>`, `<MONEY>`) using an OrderedDict to enforce conflict-free processing order. The pipeline executes in the following order:

- **Platform Links:** Zalo and Telegram links (`<LINK>`).
- **Emails, URLs:** Standard email regex; custom logic for obfuscated malicious links (`<EMAIL>`, `<URL>`).
- **Bank Accounts, Telecommunications:** Context-aware extraction for hotlines, landlines, and mobile numbers.
- **Datetime, Monetary Values, Codes & OTPs:** `<TIME>`, `<MONEY>`, `<CODE>`.
- **Anti-Evasion URL Detection:** Detects intentional spacing in domains (e.g. bit. ly/abc), and protocol-agnostic domains matching high-risk TLD patterns (.xyz, .top, .moe).

b) Layer 2 - Text Normalization and Leetspeak Decoding.: The *TextNormalizer* standardizes the character set and decodes obfuscations via a weighted scoring system validated against a Vietnamese dictionary. Three levels of obfuscation are recognized:

- **Symbol Leet** (weight 1.2–1.5): e.g. @→a, !→i. Highly indicative of spam.
- **Visual Leet** (weight 1.0): e.g. 0→o, 4→a (*h0tro* → *ho tro*).
- **Teencode** (weight 0.3): e.g. *ko*→*khong*.

A placeholder mechanism protects Layer 1 tokens from corruption during separator removal.

c) Layer 3 - Whitelist Filtering.: Valid out-of-dictionary tokens - brand names (Shopee, VCB), technical jargon (OTP, 4G), and SMS abbreviations (*lh*, *sdt*) - are exempted from further processing to prevent erroneous spell-check corrections (e.g. "VCB" → "viec").

d) Layer 4 - Spell Check and Anomaly Detection.: Remaining Out-Of-Vocabulary (OOV) tokens are analyzed by a Dual-Lookup Mechanism (full-diacritics dictionary + accent-free shadow dictionary). Tokens failing both checks are tagged as one of four error categories used as classifier features: *Telex Artifacts* (aa, aw, dd), *Run-On Words* (bisection heuristic, e.g. *vui + long*), *Gibberish Strings* (violating Vietnamese phonotactics), or *Character Repetition* (3+ identical characters, e.g. *HOTTT*). A Data Capacity Whitelist proactively exempts telco tokens (e.g. `3gb`, `12mbps`) that are technically OOV but valid in SMS contexts.

V. PROPOSED SYSTEM METHODOLOGY

The framework combines a Machine Learning scorer, a Domain Verification module, and a Hybrid Decision Logic. A purely statistical model is vulnerable to adversarial evasion; a purely rule-based system cannot generalize to novel attack patterns. The hybrid architecture addresses both shortcomings.

A. Machine Learning Model

The raw SMS T is passed through the 4-Layer Pipeline (Section IV) to generate a 27-dimensional linguistic vector F_{vec}. Unlike sparse TF-IDF matrices, this dense vector directly encodes obfuscation indicators prevalent in Vietnamese fraud. The Sender Type S (Brandname, Phone Number, etc.) is label-encoded and appended:

$$V = \text{COMBINE}(F_{\text{vec}}, S) \in \mathbb{R}^{28} \tag{1}$$

The pre-trained XGBoost model M_{AI} produces a probability score:

$$P_{AI} = M_{AI}.\text{predict\_proba}(V), \quad P_{AI} \in [0, 1] \tag{2}$$

A score near 1 reflects high obfuscation and urgency cues; near 0 indicates legitimate patterns. P_{AI} feeds the Hybrid Decision Logic.

B. Domain Verification

The Domain Verification module provides deterministic, AI-independent verdicts for URL-bearing messages (Fig. 2). The brand context B and all URLs are extracted; `tldextract` isolates the registered domain D to prevent subdomain spoofing (e.g. distinguishing *vietcombank.ngrok.io* as belonging to *ngrok.io*). If $D \in W_{\text{static}}$ (a pre-compiled whitelist of major Vietnamese banks, telcos, and global platforms), the domain is flagged Legitimate (Risk=-1.0). Otherwise, two-level dynamic verification triggers: Level 1 checks whether D appears in the top-5 search results for the brand name; Level 2 scans the site's source for internal links pointing to verified domains. An unreachable, redirect-heavy, or unverifiable site yields Risk=1.0.

C. Hybrid Decision Logic

The Hybrid Decision Logic applies a three-tier waterfall (Algorithm 1; system architecture in Fig. 3).

Tier 1 - Domain-Driven Verdict: If Risk ≥ 0.8, the message is immediately classified as Smishing. For trusted domains (Risk=-1.0), user-generated-content platforms undergo a secondary AI check (P_{AI}>0.65).

Tier 2 - Context-Aware AI Verdict: When no URL is present or domain status is inconclusive, P_{AI} is evaluated alongside context flags. Conversational messages override high AI risk;

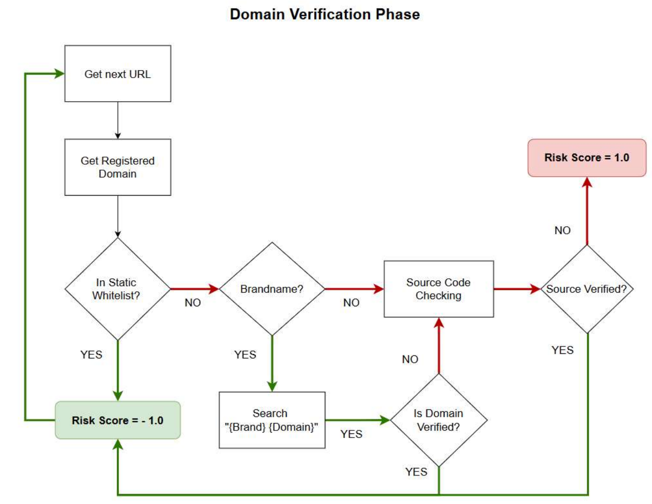

Fig. 2. Architecture of the Domain Verification Phase.

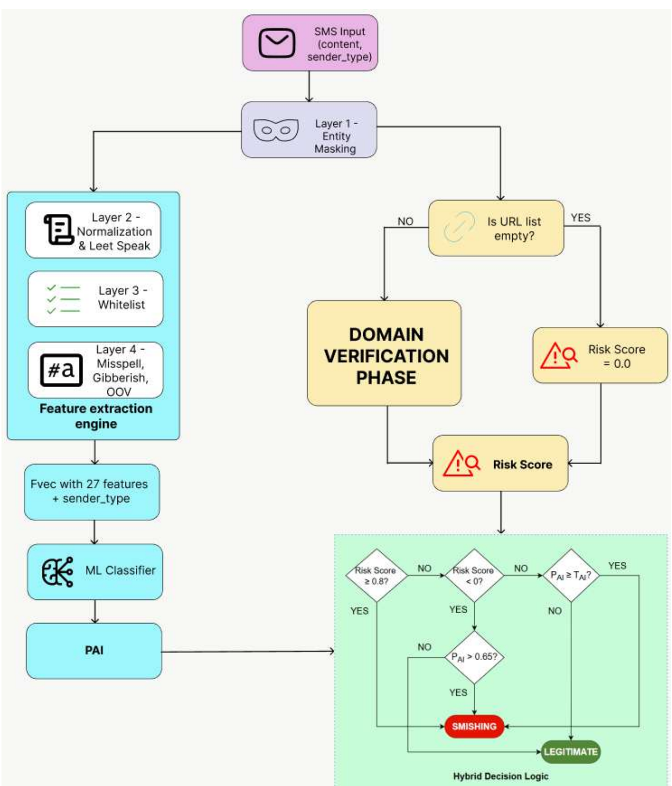

Fig. 3. Architecture of the proposed Hybrid Smishing Detection System.

messages with danger keywords from unknown senders force a Smishing verdict.

Tier 3 - Explainable Output: Every decision yields a tuple (Label, Score, Decision Phase, Reasoning) for user and analyst transparency.

Algorithm 1 Hybrid Smishing Detection Algorithm

Require: Incoming SMS T, Sender Type S, Thresholds T_{AI}, T_{high}
Ensure: Label (*Smishing / Legitimate*)

```
F_vec ← ExtractFeatures(T) {Layers 1–4}
V ← Combine(F_vec, S)
P_AI ← M_AI.predict_proba(V)
Risk ← 0
if T contains URL then
    Risk ← VerifyDomain(T)
end if
if Risk ≥ T_high then
    return Smishing {Tier 1: High-Risk Domain}
else if Risk = −1.0 then
    if P_AI > 0.65 then
        return Smishing (Warning) {UGC Exception}
    else
        return Legitimate
    end if
else
    if P_AI ≥ T_AI then
        return Smishing
    else
        return Legitimate
    end if
end if
```

VI. Experiments

A. Methods and Experimental Setup

The 2,600-message dataset (90% Legitimate, 10% Smishing) was split 80/20 using stratified sampling (`random_state=42, stratify=y`), yielding a test set of 520 messages (465 Legitimate, 55 Smishing). We adopt a two-level evaluation strategy: (i) three standalone classifiers evaluated on the full 520-sample test set using 28-dimensional features, and (ii) the complete Hybrid System evaluated on 100 stratified messages with per-error categorization.

B. Evaluation Metrics

All Precision, Recall, and F1 values refer to the *Smishing* minority class, the operationally relevant class in cybersecurity deployment. Metrics are reported on $[0, 1]$. Rather than a fixed threshold of 0.5, each classifier's optimal threshold τ^* is selected as the maximum-F1 operating point on the Precision-Recall curve:

$$\tau^* = \arg\max_{\tau} \frac{2 \cdot P(\tau) \cdot R(\tau)}{P(\tau) + R(\tau)}.$$

The Smishing F1-Score is the primary comparison metric, as overall Accuracy can be misleading under a 90/10 class imbalance [14].

C. Experimental Results

We compare the 4-Layer Pipeline against a TF-IDF Baseline (5,000 unigrams/bigrams augmented with *has_url*, *has_phone*, *sender_type*; fixed threshold 0.5). The results are presented in Table II.

TABLE II
TF-IDF BASELINE VS. PROPOSED 4-LAYER LINGUISTIC PIPELINE

Feature Set	Model	Acc.	Prec.	Recall	F1	ROC-AUC
TF-IDF Baseline (τ=0.5)	Decision Tree	0.9231	0.6000	0.8182	0.6923	0.8831
	Random Forest	0.9673	0.9750	0.7091	0.8211	0.9844
	XGBoost	0.9673	0.8800	0.8000	0.8381	0.9696
4-Layer Linguistic (τ=τ^*)	Decision Tree	0.9519	0.7679	0.7818	0.7748	0.9050
	Random Forest	0.9654	0.9744	0.6909	0.8085	0.9730
	XGBoost	**0.9750**	**0.9773**	**0.7818**	**0.8687**	**0.9659**

Prec., Recall, F1 are for the Smishing class.

The linguistic pipeline yields the most pronounced gains for XGBoost: Smishing F1 rises from 0.838 to **0.869** (+3.1 pp), and-crucially for operational deployment-Precision rises from 0.88 to **0.977** (+9.7 pp), meaning fewer than 2.3% of flagged messages are false alarms. Decision Tree benefits most in F1 (+8.3 pp), confirming that hand-crafted obfuscation indicators carry discriminative signal that sparse TF-IDF cannot capture for leet-encoded Vietnamese text.

D. Standalone Model Results

Table III reports full results at optimal thresholds.

TABLE III
SMISHING-CLASS METRICS AT OPTIMAL THRESHOLD (4-LAYER LINGUISTIC FEATURES; TEST SET n=520)

Model	τ^*	Acc.	Prec.	Recall	F1	TP	FP
Decision Tree	0.8942	0.9519	0.7679	0.7818	0.7748	43	13
Random Forest	0.6842	0.9654	0.9744	0.6909	0.8085	38	1
XGBoost	**0.8336**	**0.9750**	**0.9773**	**0.7818**	**0.8687**	**43**	**1**

FN = 55 − TP; TN = 465 − FP.

Across all three models, there is a trade-off between Precision and Recall that reflects a common challenge in imbalanced classification. Decision Tree achieves the highest Recall (0.782) but incurs 13 false positives, while Random Forest holds Precision at 0.974 at the cost of missing 17 smishing messages (Recall = 0.691). XGBoost balances both criteria, matching Decision Tree's Recall while reducing false positives to 1 - yielding the highest Smishing F1 (0.869) and overall Accuracy (0.975) among the three classifiers.

VII. RESULT ANALYSIS

A. Standalone Model Analysis

Decision Tree: The Decision Tree detects 43 of 55 smishing messages (Recall = 0.78), but at the cost of 13 false positives - roughly 1 in 4 flagged messages is actually benign (Precision = 0.768), a high false-alarm rate for production use. Overall accuracy is 95.2%.

Random Forest: Random Forest produces only 1 false positive across 465 legitimate messages (Precision = 0.974), but misses 17 smishing messages (Recall = 0.69; 31% miss rate). In a security context, missing nearly one in three attacks is a notable drawback. Random Forest may be better suited to scenarios where false alarms carry higher cost than missed attacks.

XGBoost: XGBoost achieves a balanced trade-off for cybersecurity deployment. It detects the same 43 of 55 smishing messages as Decision Tree while producing only 1 false positive - catching 5 more attacks than Random Forest (+8.3 pp Recall) at equal false-alarm cost. Its Smishing F1 of 0.8687 is the highest among the three, and its overall Accuracy of 0.9750 reflects consistent performance on both classes. XGBoost is therefore selected as the core classifier of the Hybrid Detection Framework.

B. Error Analysis of the Full Hybrid System

The complete Hybrid System ($\tau \approx 0.46$) was applied to 100 stratified messages. Of these, 87 were correctly classified, yielding 3 False Negatives (FN) and 10 False Positives (FP), summarized in Table IV.

TABLE IV
ERROR ANALYSIS ON 100 SAMPLED MESSAGES (HYBRID SYSTEM)

Type	Category	Count
FN	No-URL / low-obfuscation smishing below AI threshold	3
FP	AI model over-triggered on transactional messages	5
FP	Legitimate domain absent from Domain Risk whitelist	5
Total errors		**13**

False Negatives: All three FNs are smishing messages where neither the AI model nor the Domain Risk module exceeded their respective thresholds. Three distinct failure patterns were identified:

1) **Authority impersonation without a URL.** A message impersonating *Cảnh sát Giao thông*-containing only a hotline number-received P_{AI}=0.20 because it is written in clean, formal Vietnamese that lacks the obfuscation cues the model relies on. The Domain Risk module had no URL to evaluate.
2) **Telegram social-engineering link with minimal obfuscation.** A message linking to `t.me/hn_silent` was scored Risk=0.0 because `t.me` is whitelisted as a known messaging platform, and the AI gave P_{AI}=0.09 as the text resembles a social invitation rather than a financial scam.
3) **Leet-speak phishing URL just below the decision boundary.** A VNeID impersonation with leet-encoded text obtained P_{AI}=0.45 - one point below $\tau^*\approx$0.46- while the Domain Risk module returned r=0.0 for `vne1d-update.t0p`, finding no conclusive blacklist evidence.

These cases motivate: (a) a recall-biased rule for URL-free government/law-enforcement impersonation, and (b) a pattern-based leet-URL detector for domains whose structure matches known brand names.

False Positives: Both FP clusters arise from the system's inability to distinguish surface-level similarity from genuine fraud intent. Two distinct failure patterns were identified:

1) **AI over-triggered on transactional messages.** Five legitimate confirmation messages (restaurant reservations,

taxi bookings, e-commerce notifications) were flagged because they share structural features with smishing: brevity, a specific time reference, and an entity name (e.g. a reservation confirmation scored P_{AI}=0.94). The classifier has learned co-occurrence of {short text, temporal marker, named entity} as a fraud indicator without encoding the semantic distinction between transactional confirmations and fraudulent solicitations. Remedy: augment training data with hard-negative transactional messages and introduce a dedicated *transactional intent* feature.

2) **Legitimate domain absent from the whitelist.** Five messages were escalated by the Domain Risk module because their domains were absent from the static whitelist and the live-search verifier returned inconclusive results (Risk=1.0). Affected domains include `thuedientu.gdt.gov.vn` (official tax portal), `ghn.vn` (GHN Express), and `shopeepay.vn`. Remedy: extend the whitelist to cover all `*.gov.vn` and `*.edu.vn` subdomains, and apply a low-risk prior for government and academic TLDs in the live-search verifier.

These cases motivate: (a) richer semantic features to separate transactional from fraudulent intent in the AI tier, and (b) a broader, programmatically maintained domain whitelist that covers official government and logistics namespaces.

VIII. Conclusion and Future Work

This paper presented a Hybrid Smishing Detection Framework purpose-built for the Vietnamese linguistic environment, with three primary contributions. First, we constructed a Vietnamese smishing dataset of 2,600 labeled SMS messages via a dual-channel acquisition strategy — manual transcription from community anti-scam sources and real-device extraction via mobile backup utilities — providing a reproducible benchmark with a realistic 90/10 class imbalance. Second, we proposed a four-layer linguistic preprocessing pipeline (Entity Masking $\rightarrow$ Leet Normalization $\rightarrow$ Whitelist Filtering $\rightarrow$ Anomaly Detection) that converts obfuscated Vietnamese SMS into a 28-dimensional feature vector, raising Smishing F1 from 0.838 to 0.869 (+3.1 pp) and Precision from 0.88 to 0.977 (+9.7 pp) over a TF-IDF baseline. Third, we integrated the classifier into a three-tier Hybrid Decision Logic combining domain verification and context-aware AI scoring, achieving 97.5% overall accuracy, 78.2% Smishing Recall, and 97.7% Smishing Precision on the held-out test set, with 87% correct classifications on a real-world 100-message evaluation.

Four limitations constrain the current system: limited dataset scale (247 positives), manually maintained pipeline rules susceptible to attacker adaptation, poor coverage of URL-free authority-impersonation messages, and no support for image-based SMS content. Future work will address these gaps through LLM-driven corpus expansion toward 10,000+ samples, a BARTpho-based [15] neural text normalizer trained on ViLexNorm to replace brittle regex rules, and Rationale-Aware Knowledge Distillation into a compact ViSoBERT [16] student model targeting a Smishing F1 above 90%.

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